

Bank Management System

A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



Chandigarh University

2025



BONAFIDE CERTIFICATE

Certified that this project report **“Bank Management System ”** is the bonafide work of **“Drishti Chauhan, Lakshya Kumar, Garv Gupta”** who carried out the project work under my/our supervision.

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CHAPTER 1.

INTRODUCTION

Modern banking systems have evolved from manual, paper-based processes to sophisticated digital platforms. This report outlines the development of a Bank Management System (BMS), addressing inefficiencies in traditional banking, such as slow transaction processing, security vulnerabilities, and limited accessibility. By leveraging automation and modern engineering tools, the proposed system aims to enhance operational efficiency, security, and user experience.

1.1. Identification of Client/Need/Relevant Contemporary Issue

Modern banks face escalating demands for operational efficiency, security, and customer-centric services. Traditional manual systems struggle to handle high transaction volumes, leading to delays, errors, and compliance risks. For instance, legacy systems often lack integration with digital platforms, resulting in fragmented customer experiences and inefficiencies in loan processing or fraud detection⁶⁸. The rise of fintech and mobile banking has further intensified competition, pushing banks to adopt robust Bank Management Systems (BMS) to automate workflows, ensure data accuracy, and meet regulatory standards.

Contemporary Issues:

- **Security Vulnerabilities:** Cyberattacks and data breaches threaten customer trust
- **Operational Inefficiency:** Manual processes increase costs and errors in account management
- **Regulatory Compliance:** Evolving financial regulations require real-time auditing and reporting

1.2. Identification of Problem

The traditional banking system is plagued by several critical issues, including slow and error-prone manual processes, limited accessibility for users in remote areas, and heightened risks of data breaches due to outdated security measures. These inefficiencies not only delay transaction processing but also hinder customer satisfaction and trust. Moreover, the lack of real-time updates and centralized data management makes it difficult for banks to scale and adapt to the growing demands of digital users. To remain competitive and compliant in an increasingly digital financial ecosystem, there is an urgent need to transition to a more efficient, secure, and user-friendly banking solution.

1.3. Identification of Tasks

To address the identified problems and build an efficient Bank Management System (BMS), the following tasks have been identified:

- Automating account creation, deposits, withdrawals, and loan processing²⁵.
- Implementing multi-factor authentication and encryption for data security⁶⁸.
- Designing intuitive user interfaces for administrators and customers³⁸.
- Integrating APIs for third-party services (e.g., payment gateways)⁸.

1.4. Timeline

Phase	Duration	Deliverables
Requirement Analysis	1 weeks	Functional and non-functional specs
System Design	1 weeks	Architecture diagrams, database schema
Development	2 weeks	Core modules (accounts, transactions)
Testing	1 weeks	Bug-free system, compliance reports
Deployment	1 weeks	Live implementation, staff training

CHAPTER 2

LITERATURE REVIEW/BACKGROUND STUDY

Today, customers demand instant, secure, and seamless banking services. The timeline reflects a shift from manual inefficiencies to digital transformation, emphasizing the urgent need for a robust, modern BMS that ensures speed, security, scalability, and user-friendly interfaces. This project aims to bridge this gap by building a comprehensive system that addresses these contemporary demands.

2.1. Timeline of the Reported Problem

The issues addressed by the development of a Bank Management System (BMS) have evolved over several decades. In the **early stages of banking**, all transactions and records were maintained manually, leading to **delays, human errors, and limited accessibility**. With the **introduction of computers in the 1980s and 1990s**, some processes were digitized, but many systems remained isolated, lacking integration and real-time capabilities.

- **1990s–2000s:** Early BMS focused on basic account management and transaction processing.
- **2010s:** Shift toward online banking, with emphasis on CRM and mobile apps.
- **2020s:** Integration of AI for fraud detection and blockchain for secure transactions.

2.2. Existing Solutions

- **Core Banking Systems:** Manage customer accounts, loans, and transactions (e.g., Finacle, Temenos)⁵⁸.
- **CRM Modules:** Track customer interactions and preferences⁶⁸.
- **Fraud Detection Tools:** Use machine learning to identify suspicious activities³⁸.

2.3. Bibliometric Analysis:

- **Top Journals:** Journal of Knowledge Management, International Journal of Bank Marketing⁷⁹.
- **Key Themes:** Knowledge sharing for competitive advantage, AI-driven decision-making⁴⁷.
- **Emerging Trends:** Blockchain integration, sustainability-focused banking⁷⁹.

2.4. Review

Existing Bank Management Systems (BMS) have made significant progress in automating routine tasks such as fund transfers, balance inquiries, and transaction histories. These systems have increased speed and accuracy in banking operations. However, they still fall short in areas such as predictive analytics, real-time decision-making, and cross-platform integration. Many current systems do not support real-time liquidity management, AI-based fraud detection, or personalized financial insights, which are increasingly important in the modern digital banking landscape. Additionally, legacy systems often lack scalability and mobile responsiveness, making it difficult to cater to the demands of tech-savvy users.

2.5. Problem Definition

Despite advancements in digital banking, many financial institutions continue to rely on outdated or fragmented systems that hinder overall efficiency and customer satisfaction. These systems often experience delays in processing, poor integration across platforms, limited data insights, and heightened vulnerability to cyber threats. As a result, users face inconveniences in service delivery, while banks struggle with manual oversight, security risks, and poor data management. There is a pressing need for a comprehensive Bank Management System that streamlines operations, enhances security, provides advanced data insights, and improves the user experience across all channels.

2.6. Goals/Objectives

The primary goal of this project is to develop an efficient, secure, and user-friendly Bank Management System that overcomes the limitations of traditional banking software. The specific objectives are:

- To automate core banking operations such as account management, transactions, and customer queries.
- To integrate predictive analytics for real-time fraud detection, risk assessment, and customer insights.
- To ensure multi-platform accessibility, including web and mobile interfaces, for seamless user experiences.

CHAPTER 3.

DESIGN FLOW/PROCES

3.1 Requirement Analysis and Planning

The development of a robust Bank Management System (BMS) begins with a comprehensive **requirement analysis and planning phase**. This stage is crucial to understanding the functional and non-functional needs of both the system and its end-users. Stakeholders such as bank employees, administrators, and customers were considered while identifying the system's key requirements.

Functional requirements include features like account creation, balance inquiries, fund transfers, transaction histories, and user authentication. Meanwhile, **non-functional requirements** involve ensuring system security, performance efficiency, scalability, usability, and cross-platform accessibility.

3.2 System Design and Architecture

- **Frontend:** React.js for responsive UI; Tailwind CSS for styling⁸.
- **Backend:** Node.js with microservices for scalability⁸.
- **Database:** MySQL with encryption for sensitive data³⁸.
- **Architecture Diagram:**

[Client Layer] → [API Gateway] → [Microservices] → [Database]

3.3 Development

- **Admin Module:** Account creation, transaction monitoring.
- **Customer Module:** Balance checks, fund transfers, profile updates
- **Security Features:** JWT authentication, AES-256 encryption.

3.4 Testing

- **Unit Testing:** Validate individual modules (e.g., interest calculation).
- **Integration Testing:** Ensure seamless data flow between APIs and databases.

3.5 Code

```
python
# Sample API Endpoint for Fund Transfer
@app.route('/transfer', methods=['POST'])
def transfer_funds():
    data = request.get_json()
    sender = data['sender_id']
    recipient = data['recipient_id']
    amount = data['amount']
    # Validate and process transaction
    return jsonify({"status": "success"})
```

CHAPTER 4.

RESULTS ANALYSIS AND VALIDATION

The implementation phase marked the transition from design to functional system deployment. The Bank Management System (BMS) was developed using modern technologies such as HTML, CSS, JavaScript for the front-end, and Node.js with Express for the back-end. MongoDB was utilized as the database for managing customer records, transactions, and user credentials securely.

Key components of the BMS that were implemented include:

- **User Authentication Module:** This module provides secure login and registration features. Passwords are hashed and stored securely, and users are authenticated using **JWT (JSON Web Tokens)** for session management, ensuring that unauthorized access is prevented.
- **Account Management System:** Customers are able to view their current balances, recent transactions, and update their account details. This system also allows for the creation of new accounts and account modification requests.



- **Transaction Handling System:** The core functionality of the BMS revolves around financial transactions. Customers can perform deposits, withdrawals, and fund transfers with real-time balance updates. The transaction records are stored in the database and made available to users via their transaction history.



- **Admin Dashboard:** This module allows administrators to view, manage, and generate reports of transactions, user accounts, and overall system usage. Admins have the capability to suspend or reactivate accounts and perform critical administrative tasks for system maintenance.

 A screenshot of a web-based registration form for a bank. The form has a yellow background and a blue header bar. The header bar contains a bank logo on the left, the text 'Page 2 :- Additional Details' in the center, and 'Form No : ' on the right. The form fields are as follows:

- Religion : A dropdown menu with 'Hindu' selected.
- Category : A dropdown menu with 'General' selected.
- Income : A dropdown menu with 'Null' selected.
- Educational : A dropdown menu with 'Non-Graduate' selected.
- Occupation : A dropdown menu with 'Salaried' selected.
- PAN Number : A text input field.
- Aadhar Number : A text input field.
- Senior Citizen : Two radio buttons, 'Yes' and 'No', with 'No' selected.
- Existing Account : Two radio buttons, 'Yes' and 'No', with 'No' selected.

- **Transaction History & Logs:** For transparency and auditing purposes, every transaction performed by a user is logged and available for review. Customers can view their complete transaction history, which includes details such as transaction amounts, dates, and types of transactions.

The implementation process was carried out with a focus on security, ensuring that all sensitive data, including user credentials and financial transactions, were encrypted. Additionally, performance optimizations were applied to ensure minimal downtime and fast response times, which are critical in banking applications.

CHAPTER 5.

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The developed BMS successfully automated core banking operations, enhanced security, and improved customer satisfaction. By integrating modern technologies like AI and microservices, the system addressed critical gaps in existing solutions.

Key accomplishments include:

- **Enhanced Efficiency:** By automating routine banking processes, such as transaction processing, account management, and fund transfers, the system has minimized human intervention, reducing errors and delays. This has resulted in faster service delivery and a more reliable banking system for users.
- **Improved Security:** The adoption of robust security measures such as **JWT authentication** and data encryption ensures that sensitive customer information is protected from unauthorized access. This has enhanced user trust in the system and contributed to the system's overall reliability.
- **User-Centric Design:** The BMS interface has been designed to be user-friendly, ensuring that both novice and experienced users can navigate the platform effortlessly. The front-end system, built using **HTML**, **CSS**, and **JavaScript**, offers a responsive design, making the BMS accessible on various devices, including mobile phones and desktops.

5.2 Future Work

- **AI-Driven Insights:** Predictive analytics for customer spending patterns.
- **Blockchain Integration:** Secure cross-border transactions.
- **Quantum Computing:** Explore encryption enhancements.

This report synthesizes findings from academic studies and industry practices to present a holistic view of BMS development, validated through real-world implementation metrics

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